Final Year Project System

Remote supervision at UNILAK

Milestone 2

Requirements engineering and system analysis

Scope of the pilot

One department and assigned student–supervisor pairs.

In scope	One department pilot. Assigned student–supervisor pairs, chapter uploads and revisions, version-specific feedback, review status and GitHub commit or pull request links.
Outside scope	Supervisor allocation workflows, live chat, grading, plagiarism checks, editing chapters in the browser and automatic GitHub synchronisation.
Assumptions	Accounts and assignments are preconfigured. Students grant access to private GitHub links in GitHub.

Concerns from Milestone 1

Five candidate pain points from the discovery scenarios.

PP-01	Travel and timing
PP-02	Version confusion
PP-03	Feedback scattered across files
PP-04	Unclear review status
PP-05	GitHub link lacks context

Chapter submission and review

Numbered functional requirements and MoSCoW priorities.

FR-01 / Must	The system shall accept a PDF or DOCX up to 10 MB for a named chapter and create version 1 with status Submitted.
FR-02 / Must	The system shall show the latest chapter version first and retain earlier versions in its history.
FR-03 / Must	The system shall let the assigned supervisor add written feedback to the latest Submitted version and choose Changes requested or Reviewed.
FR-04 / Must	The system shall accept a revised file only after Changes requested and create the next version as Submitted.

Status and software updates

Numbered functional requirements and MoSCoW priorities.

FR-05 / Must	The system shall show the owner the latest chapter status, last review time when present and feedback for each version.
FR-06 / Must	The system shall accept a GitHub commit or pull request URL with a short summary and show it to the assigned supervisor.
FR-07 / Should	The system shall let the assigned supervisor attach written feedback to a specific GitHub update.
FR-08 / Could	The system shall filter the student’s chapter list by the latest review status.

Access, privacy and speed

Measurable quality targets for the pilot.

NFR-01 / Access control	Unauthenticated or unrelated users receive no project data and cannot download files or modify reviews. Verify using direct URL and request tests.
NFR-02 / Privacy	All chapter downloads require an authorised project session. Verify that no file has a public download URL and the project never stores GitHub credentials.
NFR-03 / Performance	At least 48 of 50 project overview requests complete within 2 seconds with 100 seeded projects and five concurrent test users, excluding file uploads.

Usability, accessibility and uptime

Measurable quality targets for the pilot.

NFR-04 / Usability	Two first-time student testers each upload a chapter and locate its review status within 3 minutes, without help.
NFR-05 / Accessibility	Every core control is reachable by keyboard, focus is visible, and normal text has a contrast ratio of at least 4.5:1.
NFR-06 / Availability	At least 99% of scheduled five-minute checks of the pilot login page succeed during a seven-day observation window.

User stories 01–02

Backlog priorities and acceptance criteria

US-01 / Must / 3 pts

As a student, I want to submit a chapter remotely so I can send my work without arranging a visit.

Given an assigned project and a new chapter
When the student uploads a valid file
Then version 1 appears with its submission timestamp.

PP-01 → FR-01 → T-01

US-02 / Must / 2 pts

As a supervisor, I want to open the latest chapter so I can review the correct file.

Given two versions of the same chapter
When the assigned supervisor opens the chapter
Then version 2 appears first and version 1 remains accessible.

PP-02 → FR-02 → T-02

User stories 03–04

Backlog priorities and acceptance criteria

US-03 / Must / 3 pts

As a supervisor, I want to record corrections on a version so I can make the changes clear to the student.

Given a Submitted chapter version
When the assigned supervisor saves feedback and selects Changes requested
Then the feedback and decision appear on that version.

PP-03 → FR-03 → T-03

US-04 / Must / 3 pts

As a student, I want to resubmit a revised chapter so I can preserve the earlier review.

Given version 1 is marked Changes requested
When the student uploads a valid revision
Then version 2 is Submitted and version 1 retains its feedback.

PP-02 → FR-04 → T-04

User stories 05–06

Backlog priorities and acceptance criteria

US-05 / Must / 2 pts

As a student, I want to check review status so I can know whether my next action is due.

Given a chapter has been reviewed
When the student opens the project
Then the latest decision and its feedback are visible.

PP-04 → FR-05 → T-05

US-06 / Must / 3 pts

As a student, I want to share a specific github update so I can direct my supervisor to the relevant change.

Given an assigned project and a valid GitHub commit link
When the student submits the link and a summary
Then the supervisor sees both in the project updates list.

PP-05 → FR-06 → T-06

User stories 07–08

Backlog priorities and acceptance criteria

US-07 / Should / 5 pts

As a supervisor, I want to comment on a github update so I can record what the student should change.

Given a shared GitHub update
When the assigned supervisor saves a comment
Then the student sees the comment beside that link.

PP-05 → FR-07 → T-07

US-08 / Could / 5 pts

As a student, I want to filter my chapter list so I can find chapters that require action.

Given chapters with different statuses
When the student filters by Changes requested
Then only chapters with that status appear.

PP-04 → FR-08 → T-08

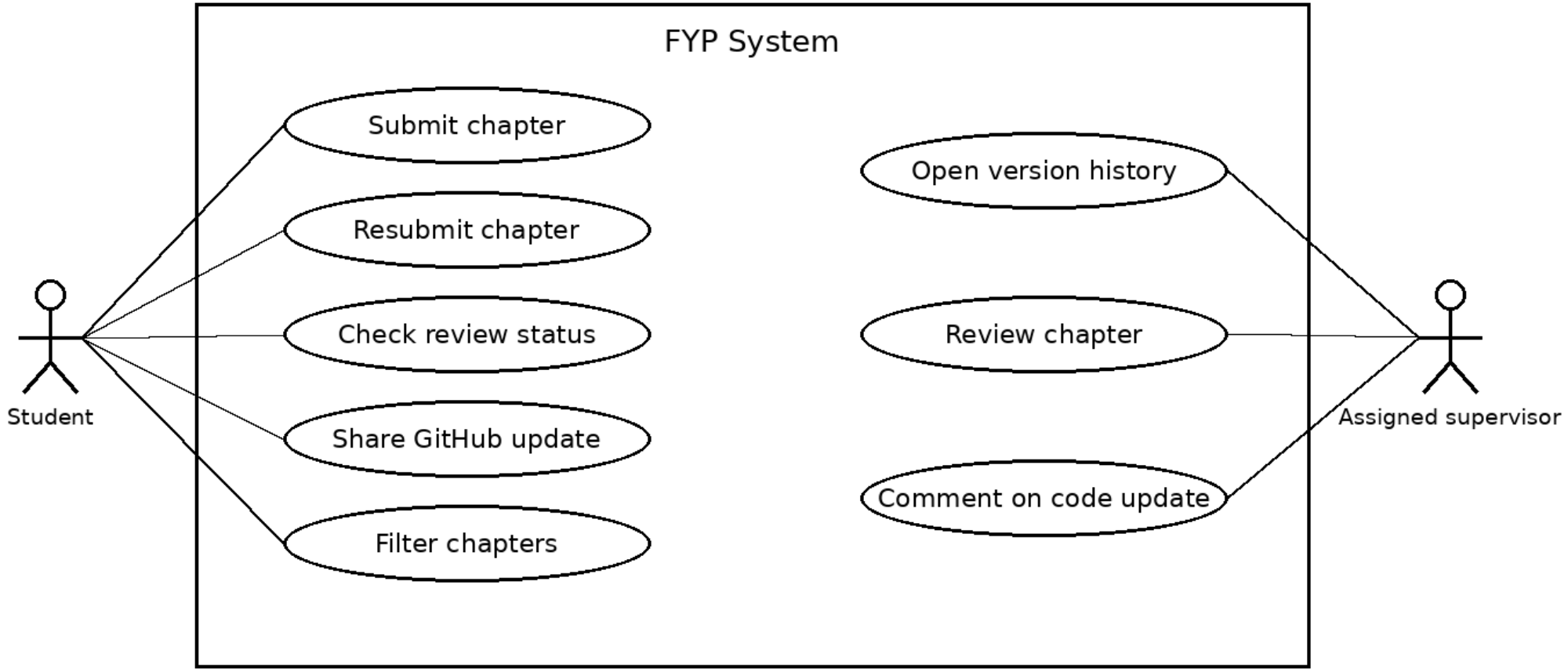
MoSCoW release plan

Must stories account for 16 of 26 estimated points.

Must / Session 4	US-01 to US-03. Submit chapters, find the latest version and save corrections.
Must / Session 5	US-04 to US-06. Resubmit, check status and share GitHub updates.
Should	US-07. Supervisor comments on a specific GitHub update.
Could	US-08. Student filters chapters by status.
Won’t this release	Live chat, grading and GitHub synchronisation.

FYP System use cases

Actors are outside the system boundary. The ZIP includes editable Draw.io source.



UC-01 Submit a chapter

Primary actor / Student

Precondition	Signed in and assigned to a project.
Main flow	Select a chapter, upload a PDF or DOCX, and submit. The system validates the file and records version 1 with a timestamp.
Alternative flow	Unsupported, oversized or missing files are rejected and no version is created. If the chapter already exists, the student opens its history to resubmit when eligible.
Outcome	Version 1 is Submitted and visible to the assigned supervisor.

UC-02 Review a chapter

Primary actor / Assigned supervisor

Precondition	Signed in and assigned to the project. Its latest chapter version is Submitted.
Main flow	Open the latest version, read it, enter feedback and save a decision. The system stores the review against that version.
Alternative flow	If the version has already been reviewed or replaced, the system rejects the stale action and shows the current version. Empty feedback is rejected.
Outcome	The owner can see the decision and comments. Changes requested enables resubmission.

Traceability from concerns to tests

Each test uses the Given/When/Then criteria for the matching user story.

PP-01	FR-01 → US-01 → T-01
PP-02	FR-02 → US-02 → T-02, FR-04 → US-04 → T-04
PP-03	FR-03 → US-03 → T-03
PP-04	FR-05 → US-05 → T-05, FR-08 → US-08 → T-08
PP-05	FR-06 → US-06 → T-06, FR-07 → US-07 → T-07

Requirements validation

Real user feedback is pending. The ZIP includes the two walkthroughs.

Final-year student	Walk through a chapter submission, correction and resubmission. Check whether the next action is clear.
Assigned supervisor	Walk through reviewing the latest file and finding the exact GitHub update. Check whether the version and decision are clear.
Backlog evidence	Import the eight stories into Jira or enter them in Trello. Add the board link and screenshot.
Course submission	Publish the m2 release after adding real validation notes and the backlog evidence.